A SCHEME IS EXCELLENT EXACTLY WHEN IT IS QUASI-EXCELLENT AND UNIVERSALLY CATENARY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-characterize-excellent is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 3439-3446 and proof span 3448-3459 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof contains the author's typo 'universally category' in place of 'universally catenary'; it is kept verbatim because the rule is to deliver the author's text as it stands, and the surrounding sentence makes the intended meaning unambiguous.

Editorial note. 1 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- Excellent rings are quasi-excellent and universally catenary. Used at proof lines 3448-3452. Gives both halves of the first implication.
- The locality statement for universal catenarity: an affine open chart of a universally catenary scheme has universally catenary coordinate ring, cited from the same chapter. Cited at proof lines 3452-3453 and 3456-3457. Converts universal catenarity of the scheme into universal catenarity of the relevant coordinate ring, and conversely.
- Quasi-excellence of a scheme means that every point has an affine open neighbourhood with quasi-excellent coordinate ring. Used at proof lines 3454-3455. Supplies the chart whose coordinate ring is quasi-excellent.

Source reference. Original lemma 02J9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3328–3328. Same-chapter reference used in the retained author text at source line(s) 3185, 3189, 3223, 3452, 3457, 3484, 13684. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let X be a scheme. The following are equivalent*

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) *X is excellent, and*
(2) *X is quasi-excellent and universally catenary.*

Proof. Assume (1). Since excellent rings are quasi-excellent, it is clear that X is quasi-excellent. Since excellent rings are universally catenary, it also follows that X is universally catenary by Lemma 02J9. Assume (2). Let $x \in X$. Since X is quasi-excellent there exists an affine open neighbourhood $x \in U \subset X$ such that $\mathcal{O}_X(U)$ is quasi-excellent. Since X is universally category, the ring $\mathcal{O}_X(U)$ is also universally catenary by Lemma 02J9, whence excellent. □